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(54) **COMMUNICATION SYSTEM, CONTROL METHOD, AND STORAGE MEDIUM**

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G06V 20/40 (2022.01)
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(Continued)

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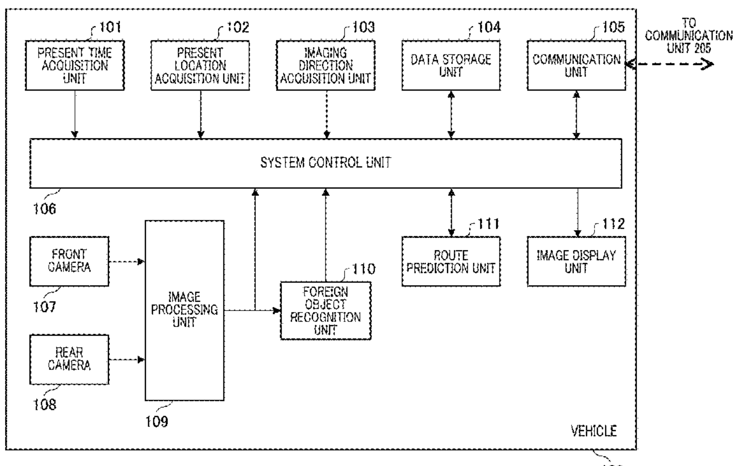
(57) **ABSTRACT**

A communication system includes a first vehicle and a second vehicle, with the first vehicle configured to capture a video and recognize a foreign object on a road from the video, and can acquire an imaging location where the video is captured, an imaging direction in which the video is captured, and transmit information. The second vehicle receives the information, acquires a traveling schedule direction of the second vehicle, and displays a video. The imaging location and the imaging direction are added to the video. The first vehicle acquires a time at which the video is captured, and the second vehicle acquires a present time, and the video is displayed when an imaging time of a presently displayed video is earlier than the present time by a given time or more despite mismatch between the acquired traveling schedule direction and the received imaging direction.

7 Claims, 8 Drawing Sheets



(51) **Int. Cl.**
G06V 20/58 (2022.01)
G08G 1/056 (2006.01)
G08G 1/16 (2006.01)

(52) **U.S. Cl.**
CPC *G08G 1/162* (2013.01); *G08G 1/164* (2013.01); *G08G 1/165* (2013.01); *G08G 1/166* (2013.01)

(58) **Field of Classification Search**
CPC G08G 1/096775; G08G 1/096791; G08G 1/162; G08G 1/164; G08G 1/165; G08G 1/166; G08G 1/0112
USPC 348/148, 159
See application file for complete search history.

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FIG. 1

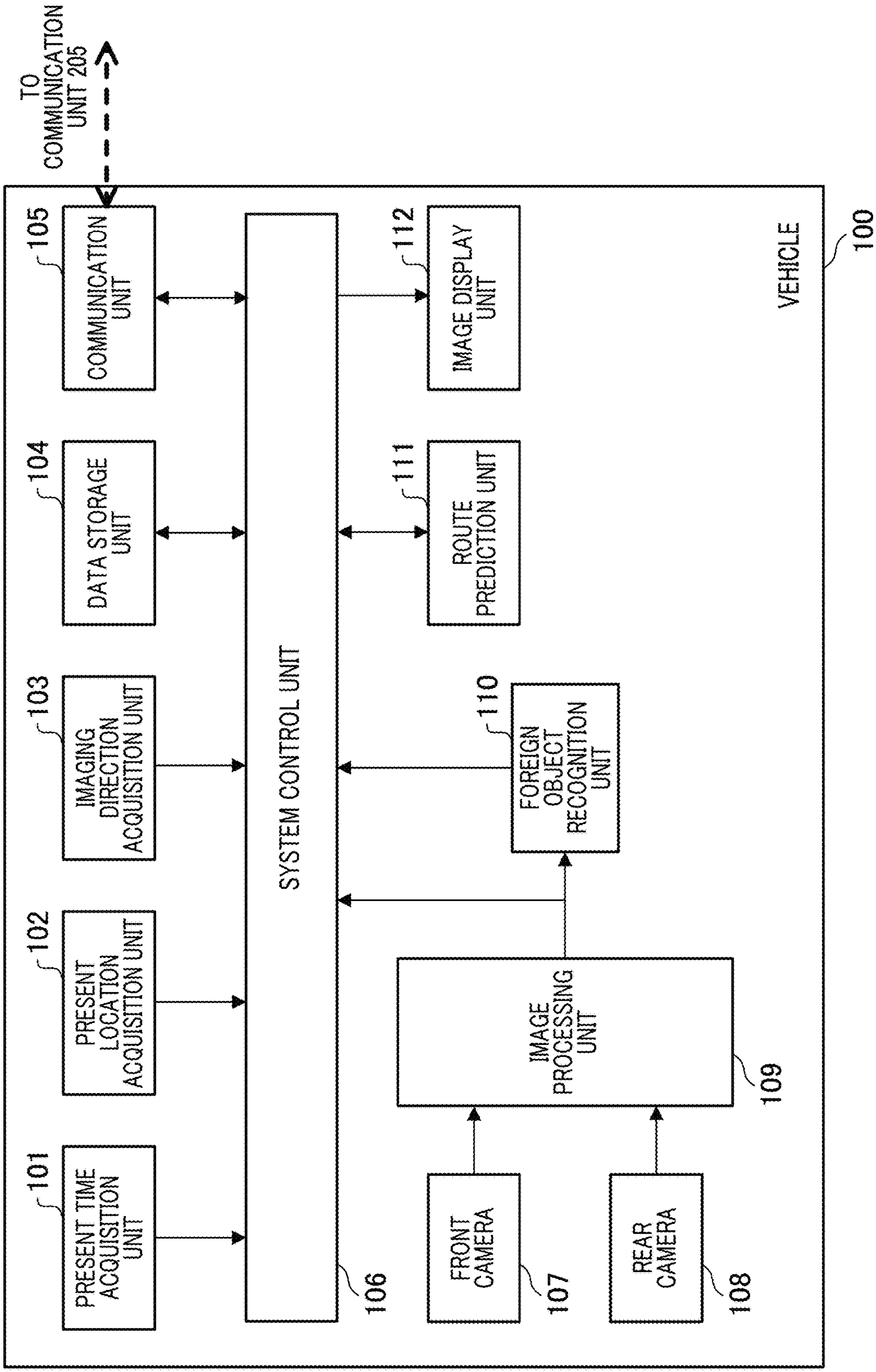

TO COMMUNICATION UNIT 205
COMMUNICATION UNIT
105
DATA STORAGE UNIT
104
IMAGING DIRECTION ACQUISITION UNIT
103
PRESENT LOCATION ACQUISITION UNIT
102
PRESENT TIME ACQUISITION UNIT
101
SYSTEM CONTROL UNIT
106
IMAGE DISPLAY UNIT
112
ROUTE PREDICTION UNIT
111
FOREIGN OBJECT RECOGNITION UNIT
110
IMAGE PROCESSING UNIT
109
FRONT CAMERA
107
REAR CAMERA
108
VEHICLE
100

FIG. 2

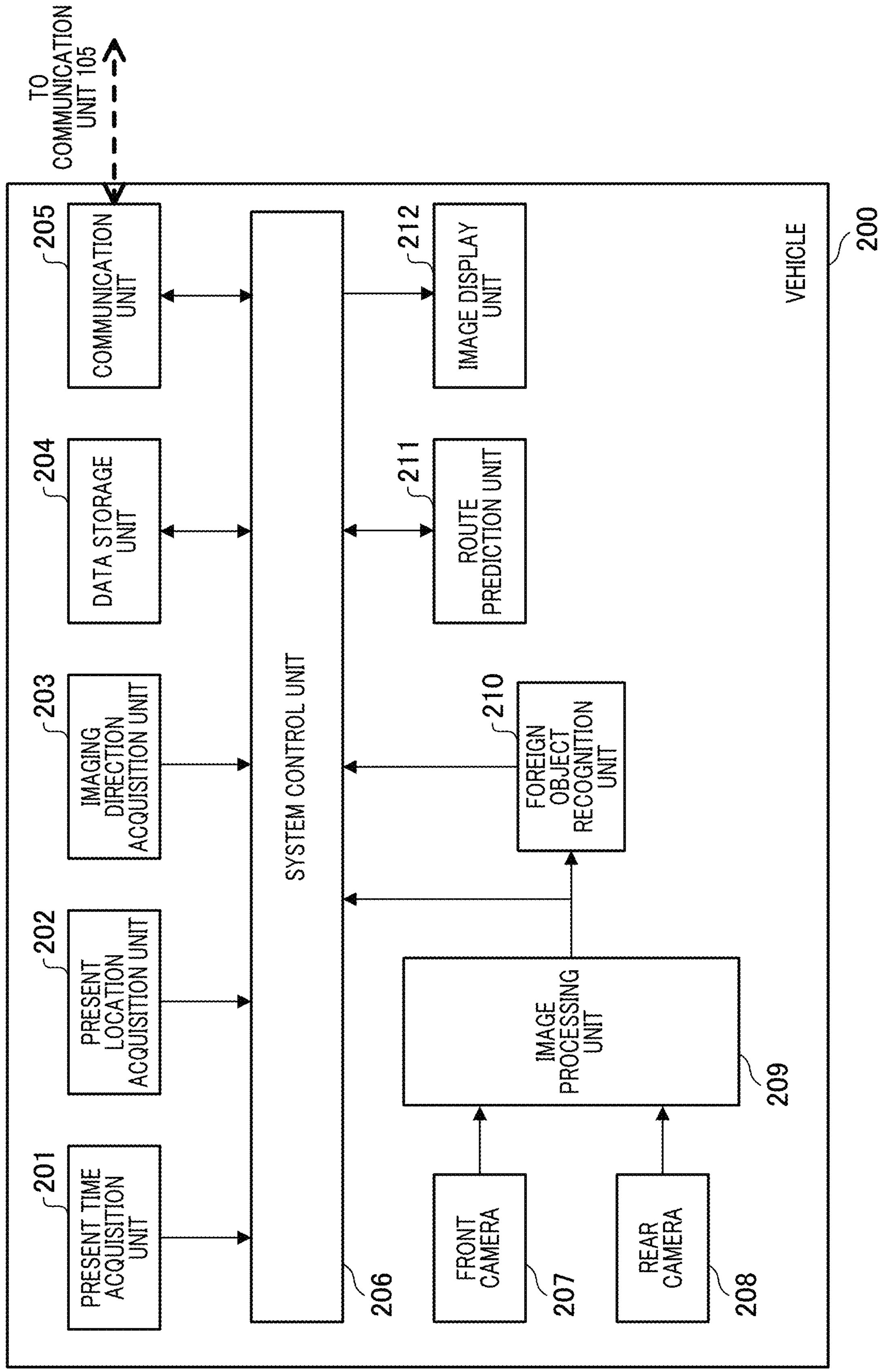

TO COMMUNICATION UNIT 105
205
COMMUNICATION UNIT
204
DATA STORAGE UNIT
203
IMAGING DIRECTION ACQUISITION UNIT
202
PRESENT LOCATION ACQUISITION UNIT
201
PRESENT TIME ACQUISITION UNIT
SYSTEM CONTROL UNIT
206
212
IMAGE DISPLAY UNIT
211
ROUTE PREDICTION UNIT
210
FOREIGN OBJECT RECOGNITION UNIT
IMAGE PROCESSING UNIT
209
FRONT CAMERA
207
REAR CAMERA
208
VEHICLE
200

FIG. 3A

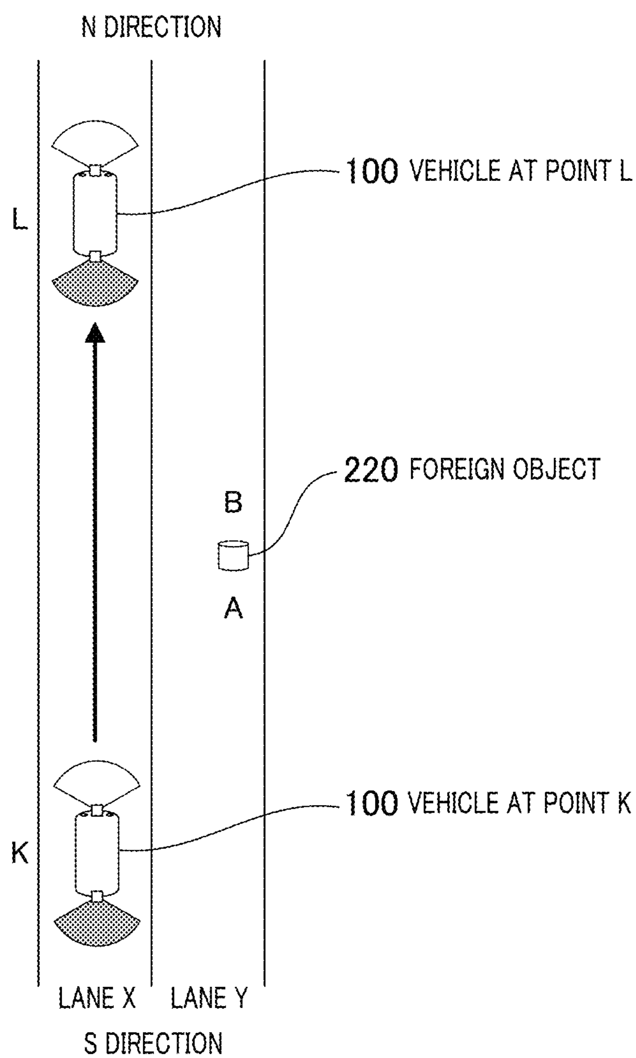

N DIRECTION
100 VEHICLE AT POINT L
L
220 FOREIGN OBJECT
B
A
100 VEHICLE AT POINT K
K
LANE X
LANE Y
S DIRECTION

FIG. 3B

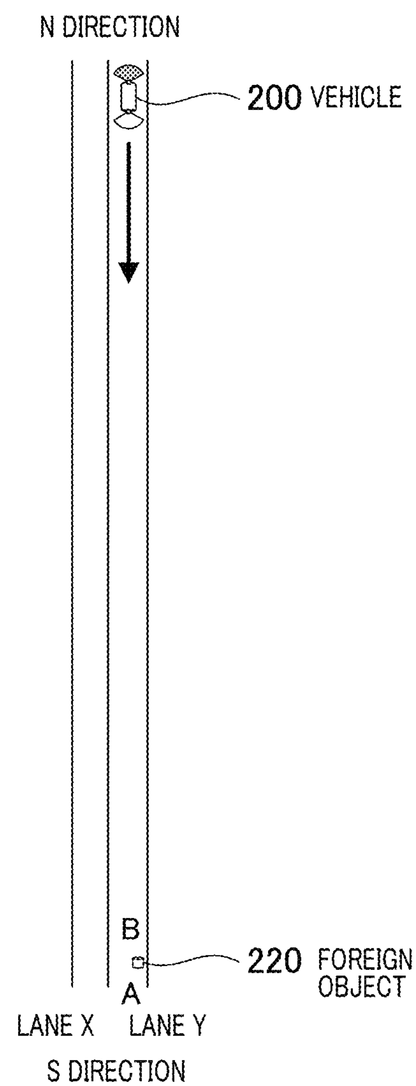

N DIRECTION
200 VEHICLE
B
220 FOREIGN OBJECT
A
LANE X
LANE Y
S DIRECTION

FIG. 4

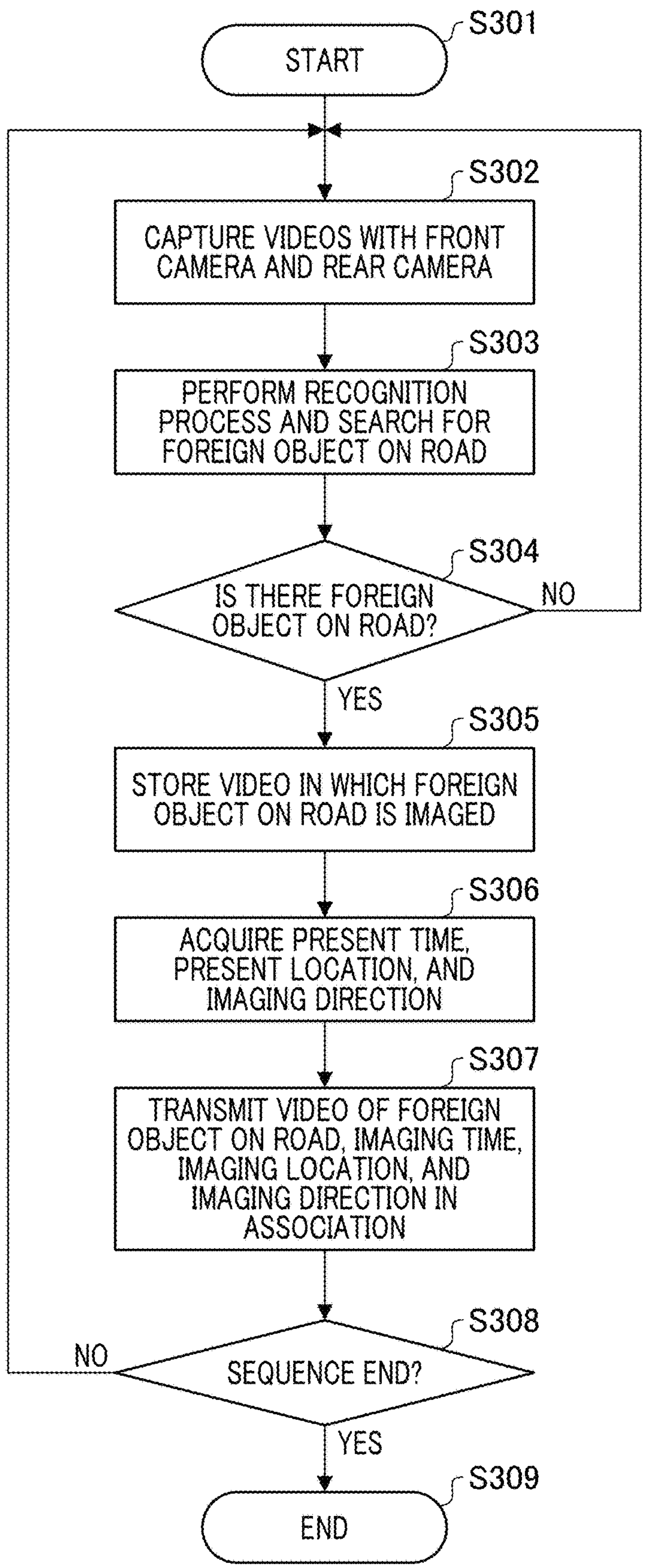
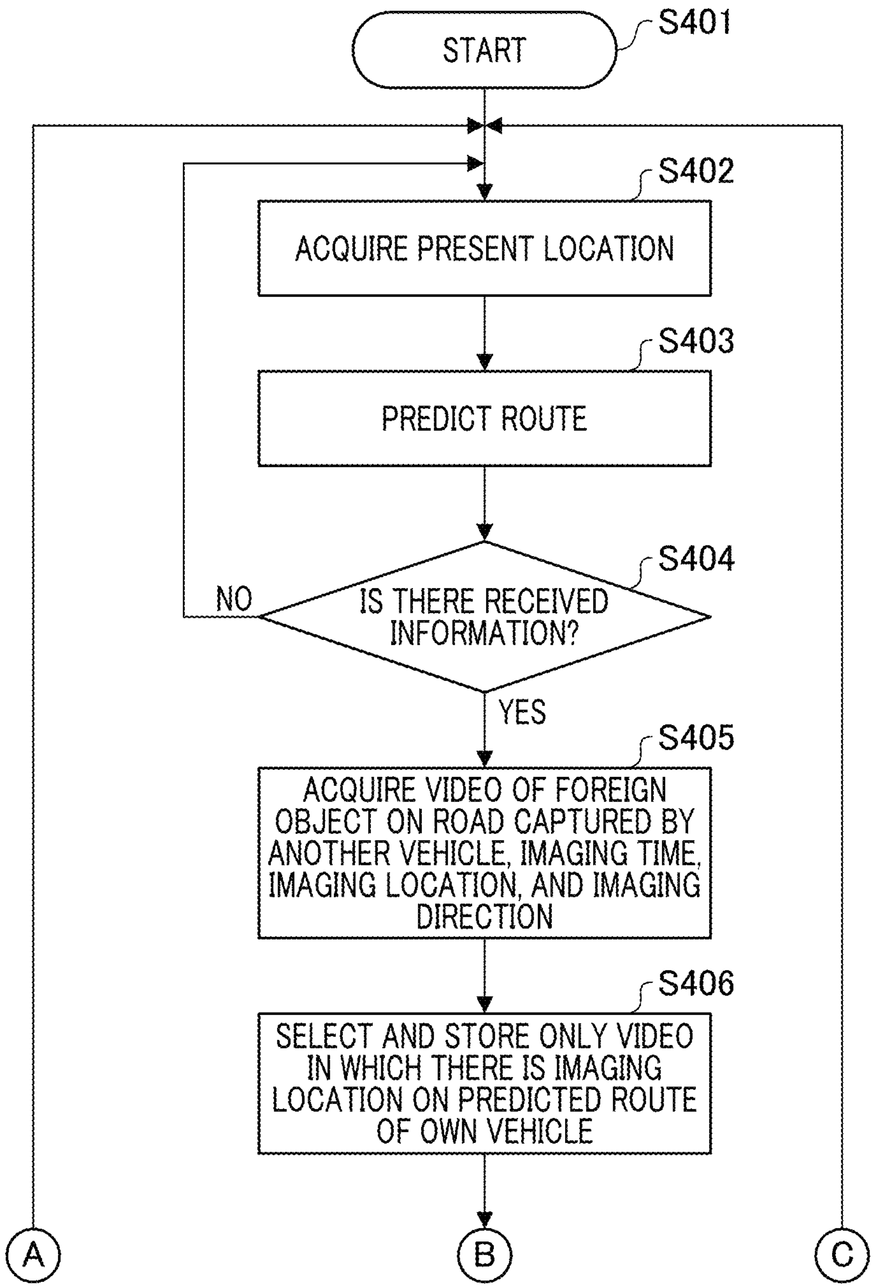

S301
START
S302
CAPTURE VIDEOS WITH FRONT CAMERA AND REAR CAMERA
S303
PERFORM RECOGNITION PROCESS AND SEARCH FOR FOREIGN OBJECT ON ROAD
S304
IS THERE FOREIGN OBJECT ON ROAD?
NO
YES
S305
STORE VIDEO IN WHICH FOREIGN OBJECT ON ROAD IS IMAGED
S306
ACQUIRE PRESENT TIME, PRESENT LOCATION, AND IMAGING DIRECTION
S307
TRANSMIT VIDEO OF FOREIGN OBJECT ON ROAD, IMAGING TIME, IMAGING LOCATION, AND IMAGING DIRECTION IN ASSOCIATION
S308
SEQUENCE END?
NO
YES
S309
END

FIG. 5



START
S401
S402
ACQUIRE PRESENT LOCATION
S403
PREDICT ROUTE
S404
IS THERE RECEIVED INFORMATION?
NO
YES
S405
ACQUIRE VIDEO OF FOREIGN OBJECT ON ROAD CAPTURED BY ANOTHER VEHICLE, IMAGING TIME, IMAGING LOCATION, AND IMAGING DIRECTION
S406
SELECT AND STORE ONLY VIDEO IN WHICH THERE IS IMAGING LOCATION ON PREDICTED ROUTE OF OWN VEHICLE
A
B
C

FIG. 6

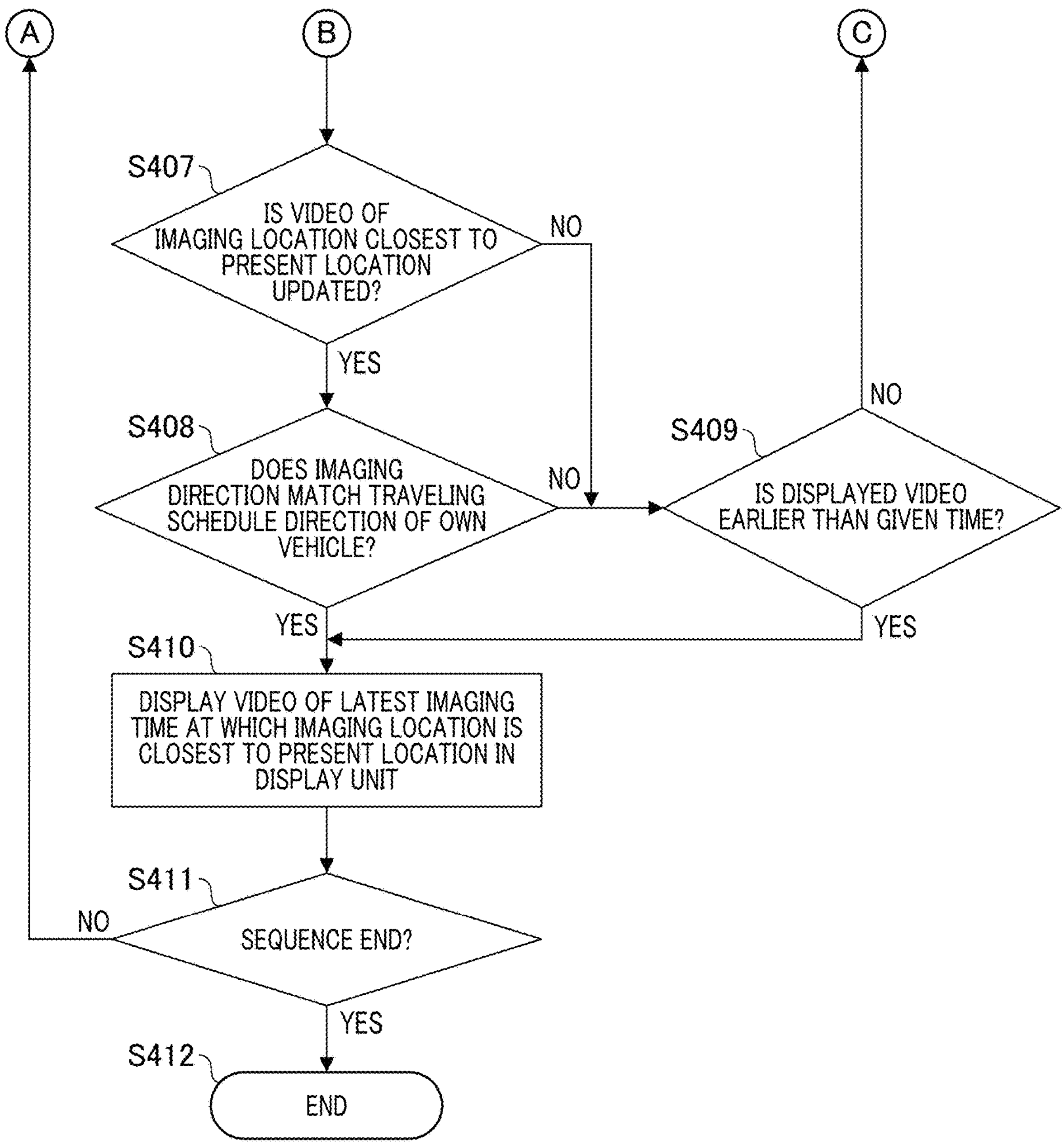

A
B
C
S407
IS VIDEO OF IMAGING LOCATION CLOSEST TO PRESENT LOCATION UPDATED?
NO
YES
S408
DOES IMAGING DIRECTION MATCH TRAVELING SCHEDULE DIRECTION OF OWN VEHICLE?
NO
YES
S409
IS DISPLAYED VIDEO EARLIER THAN GIVEN TIME?
NO
YES
S410
DISPLAY VIDEO OF LATEST IMAGING TIME AT WHICH IMAGING LOCATION IS CLOSEST TO PRESENT LOCATION IN DISPLAY UNIT
S411
SEQUENCE END?
NO
YES
S412
END

FIG. 7A

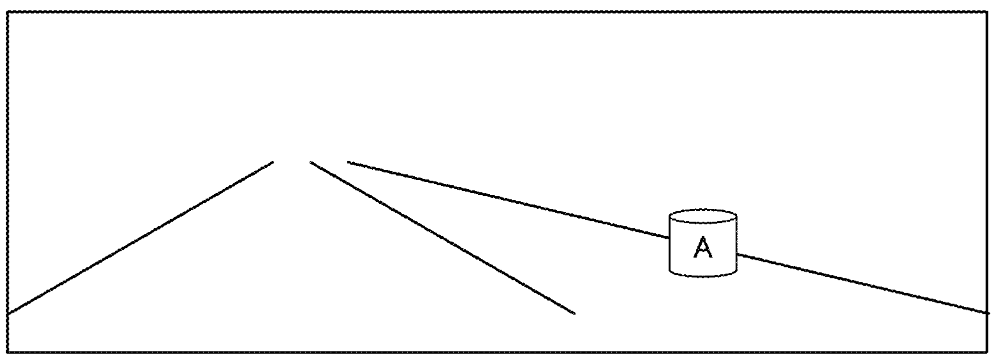


FIG. 7B

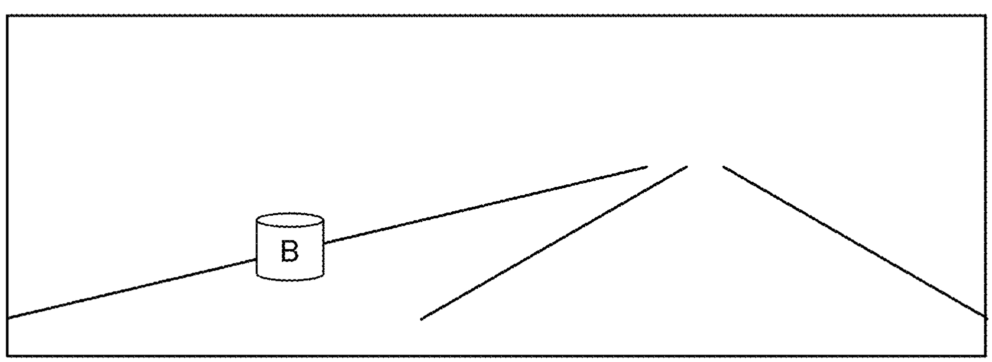


FIG. 8

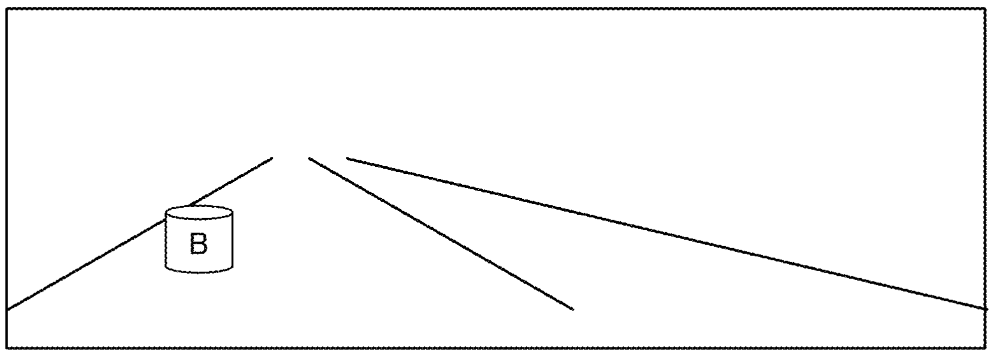
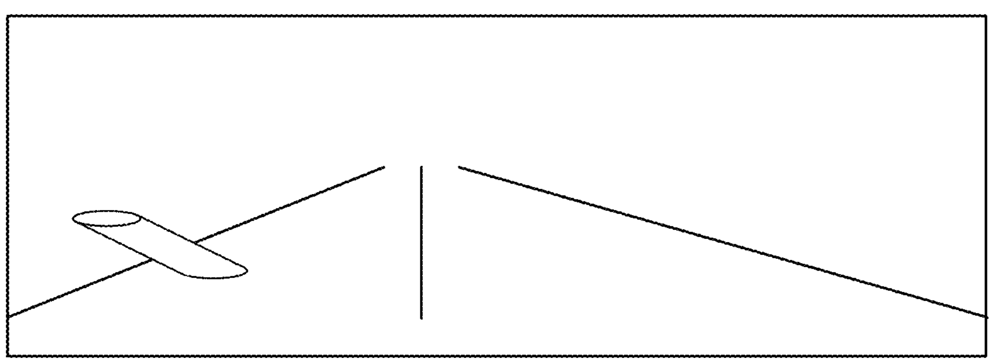


FIG. 9



COMMUNICATION SYSTEM, CONTROL METHOD, AND STORAGE MEDIUM

BACKGROUND OF THE INVENTION

Field of the Invention

The present invention relates to a communications system, a control method, and a storage medium.

Description of the Related Art

In the related art, there is a technique for transmitting a video captured with a camera mounted in a preceding vehicle to a following vehicle and displaying the video in the following vehicle. For example, Japanese Patent Laid-Open No. 2011-226972 discloses a technique for mounting a GPS receiver and a camera in a preceding train and displaying a traveling position and a video of the train on a monitor mounted in a following train. GPS is an abbreviation for Global Positioning System.

Incidentally, it is conceivable that a technique for providing a video captured in a preceding vehicle to a following vehicle be applied to a general automobile. In an automobile, the following vehicle can recognize and avoid a foreign object on a road by causing the preceding vehicle to transmit a video in which the foreign object on the road is imaged and causing the following vehicle to receive and display the video on a monitor in the vehicle.

In Japanese Patent Laid-Open No. 2011-226972, it is assumed that a following vehicle follows the same course as that of a preceding vehicle. However, there is not always a preceding vehicle in automobiles. Therefore, if there is no preceding vehicle, a video of a side in front in a traveling direction cannot be provided to an automobile.

In automobiles traveling on roads with traffic in both directions, an oncoming vehicle approaches from the side in front of an own vehicle. Therefore, the own vehicle can obtain an opportunity to acquire a video of the side in front in the traveling direction by receiving a video captured by the oncoming vehicle.

However, since a video obtained by a preceding vehicle imaging the side in front and a video obtained by an oncoming vehicle imaging the side in front are reversed in the traveling direction, there is a problem that it is difficult to understand the video when the video is used, for example, to recognize a foreign object on a road and it is difficult for a driver to take countermeasures. In this way, in the related art, there is room for an improvement in provision of a video of a side in front in a traveling direction of a vehicle.

SUMMARY OF THE INVENTION

According to an aspect of the present invention, a communication system includes a first vehicle and a second vehicle. The first vehicle includes an imaging unit configured to capture a video and at least one processor or circuit configured to function as: a foreign object recognition unit configured to recognize a foreign object on a road from the video, an imaging location acquisition unit configured to acquire an imaging location which is a location where the video is captured, an imaging direction acquisition unit configured to acquire an imaging direction which is a direction in which the video is captured, and a transmission unit configured to transmit information. The second vehicle includes at least one processor or circuit configured to function as: a reception unit configured to receive the

information, a traveling schedule direction acquisition unit configured to acquire a traveling schedule direction of the second vehicle, and a display unit configured to display a video. The transmission unit adds an imaging location acquired by the imaging location acquisition unit and an imaging direction acquired by the imaging direction acquisition unit to a video in which the foreign object recognition unit recognizes that there is a foreign object on a road and transmits the video. The reception unit receives the video in which it is recognized that there is the foreign object, the imaging location, and the imaging direction. The display unit displays a video in which the traveling schedule direction acquired by the traveling schedule direction acquisition unit matches the received imaging direction.

Further features of the present invention will become apparent from the following description of embodiments with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a block diagram illustrating a vehicle **100** according to an embodiment of the present invention.

FIG. **2** is a block diagram illustrating a vehicle **200** according to an embodiment of the present invention.

FIGS. **3**A and **3**B are diagrams illustrating a positional relation between the vehicle **100** and a foreign object **220**.

FIG. **4** is a flowchart illustrating an operation of the vehicle **100** which is a transmission side.

FIG. **5** is a flowchart illustrating an operation of the vehicle **200** which is a reception side.

FIG. **6** is a flowchart illustrating an operation of the vehicle **200** which is the reception side and a diagram continuous from FIG. **5**.

FIG. **7**A is a diagram illustrating a video of a front camera **107** of the vehicle **100** at a point K and FIG. **7**B is a diagram illustrating a video of a rear camera **108** of the vehicle **100** at a point L.

FIG. **8** is a diagram illustrating a video which can be acquired by a preceding vehicle of the vehicle **200**.

FIG. **9** is a diagram illustrating a video obtained by performing image processing on the video of FIG. **7**B.

DESCRIPTION OF THE EMBODIMENTS

Hereinafter, with reference to the accompanying drawings, favorable modes of the present invention will be described using Embodiments. In each diagram, the same reference signs are applied to the same members or elements, and duplicate description will be omitted or simplified.

Embodiment

FIGS. **1** and **2** are block diagrams illustrating vehicles including a communication system according to an embodiment of the present invention. FIG. **1** is a block diagram illustrating a vehicle **100** according to the embodiment of the present invention.

FIG. **2** is a block diagram illustrating a vehicle **200** according to the embodiment of the present invention. The communication system according to the present embodiment includes the vehicles **100** and **200**. The vehicle **100** is an example of a first vehicle. The vehicle **200** is an example of a second vehicle.

The vehicle **100** includes a present time acquisition unit **101**, a present location acquisition unit **102**, an imaging direction acquisition unit **103**, a data storage unit **104**, a communication unit **105**, and a system control unit **106**. The vehicle **100** further includes a front camera **107**, a rear camera **108**, an image processing unit **109**, a foreign object recognition unit **110**, a route prediction unit **111**, and an image display unit **112**. The vehicle **100** further includes a driving unit (not illustrated) and can be moved by own power of the driving unit.

The present time acquisition unit **101** can acquire a present time using a scheme such as Global Navigation Satellite System (GNSS). The present location acquisition unit **102** can acquire a present location using a scheme such as GNSS.

The imaging direction acquisition unit **103** can acquire azimuth information obtained by a geomagnetic sensor mounted on a vehicle body and an imaging direction of each camera provided in the vehicle body (here, the front camera **107** and the rear camera **108**)

When the geomagnetic sensor is not used, the imaging direction acquisition unit **103** may estimate a traveling direction of the vehicle **100** from a temporal change of present location information obtained from GNSS and may acquire an imaging direction of each camera from a direction of the camera provided on the vehicle body. The front camera **107** is an example of a front imaging unit that images the side in front. The rear camera **108** is an example of a rear imaging unit that images the side to the rear.

The data storage unit **104** is configured with a semiconductor memory or the like and can store acquired data. The communication unit **105** can transmit and receive data to and from another vehicle or a base station using a technique of a wireless LAN or a wireless WAN.

The system control unit **106** includes a central processing unit (CPU) that performs calculation or control, a read only memory (ROM) that is a main storage device, and a random access memory (RAM).

The system control unit **106** can receive data from the present time acquisition unit **101**, the present location acquisition unit **102**, the imaging direction acquisition unit **103**, the image processing unit **109**, and the foreign object recognition unit **110**.

The system control unit **106** can exchange data with each of the data storage unit **104**, the communication unit **105**, and the route prediction unit **111**. The system control unit **106** can transmit data to the image display unit **112**.

The front camera **107** is configured with an optical lens and a sensor and can image the side in front of the vehicle **100**. The rear camera **108** is configured with the same optical lens and sensor as those of the front camera **107** and can image the side to the rear of the vehicle **100**.

An imaging direction of the front camera **107** matches a traveling direction when the vehicle **100** moves forward. An imaging direction of the rear camera **108** matches a traveling direction when the vehicle **100** moves backward. The imaging direction of the front camera **107** and the imaging direction of the rear camera **108** are different from each other by 180°.

The image processing unit **109** performs image processing such as de-Bayer processing, distortion correction, gamma curve processing, or color space conversion on videos captured by the front camera **107** and the rear camera **108**. Accordingly, the image processing unit **109** can generate (create) videos which humans can easily identify. The image processing unit **109** can transmit the generated videos to the system control unit **106** and the foreign object recognition unit **110**.

The foreign object recognition unit **110** can perform a recognition process using a scheme such as a convolutional neural network (CNN) on the received video and can estimate a road region in the video by recognizing white lines or curbstones. The foreign object recognition unit **110** can detect a foreign object such as a fallen object, a depressed hole, or earth and sand on a road by further searching the road region.

The route prediction unit **111** can predict a route along which the vehicle **100** will travel from map information stored in the data storage unit **104**, preset location information which can be acquired from the present location acquisition unit **102**, destination information of a navigation system (not illustrated), and the like.

The image display unit **112** is configured with an LCD panel or the like and can notify a user of information by displaying a video.

The vehicle **200** also has a configuration similar to that of the vehicle **100**. The vehicle **200** includes a present time acquisition unit **201**, a present location acquisition unit **202**, an imaging direction acquisition unit **203**, a data storage unit **204**, a communication unit **205**, and a system control unit **206**.

The vehicle **200** further includes a front camera **207**, a rear camera **208**, an image processing unit **209**, a foreign object recognition unit **210**, a route prediction unit **211**, and an image display unit **212**. The vehicle **200** further includes a driving unit (not illustrated) and can be moved by own power of the driving unit.

Since an operation of each configuration of the vehicle **200** illustrated in FIG. **2** is similar to that of each configuration of the vehicle **100** illustrated in FIG. **1**, detailed description thereof will be omitted. The vehicles **100** and **200** can perform authentication using the communication units **105** and **205**, respectively, establish communication, and transmit and receive data.

Here, in description, it is assumed that the vehicle **100** is a transmission side and the vehicle **200** is a reception side, but both the vehicles **100** and **200** have both configurations of the transmission side and the reception side. However, the present system may have a vehicle with only a configuration of the transmission side or a vehicle with only a configuration of the reception side.

FIGS. **3**A and **3**B are diagrams illustrating a positional relation between the vehicle **100** and a foreign object **220**. FIG. **3**A illustrates a situation in which the vehicle **100** is moving from a point K to a point L. In FIGS. **3**A and **3**B, three vertical lanes indicate roads.

It is assumed that the left side is a lane X, the right side is a line Y, the upper side is an N direction, and the lower side is an S direction. It is assumed that a traveling direction of a vehicle in the lane X is the N direction and a traveling direction of a vehicle in the lane Y is the S direction. It is assumed that the foreign object **220** is on the lane Y, a surface of the object **220** in the S direction is an A surface and a surface of the object **220** in the N direction is a B surface.

In FIGS. **3**A and **3**B, fan-shaped portions illustrated above and below the vehicles **100** and **200** indicate imaging range angles of the front camera **207** and the rear camera **208**. A portion indicated with white in the fan-shaped portions indicates an imaging range angle of the front camera **107** and a portion indicated with gray in the fan-shaped portions indicates an imaging range angle of the rear camera **108**.

As illustrated in FIG. **3**A, the vehicle **100** is traveling in the lane X from the S direction to the N direction. The lane Y is an opposite lane of the lane X. The foreign object **220** is on the lane Y between the points K and L.

FIG. **3**B illustrates a situation in which the vehicle **200** is traveling in the lane Y from the N direction to the S direction. The vehicle **200** is located in the N direction from the point L at a time point at which the vehicle **100** reaches the point L. It is assumed that the vehicle **200** is distant from the foreign object **220** and the vehicle **200** cannot directly check the object **220**.

FIG. **4** is a flowchart illustrating an operation of the vehicle **100** which is a transmission side. FIG. **5** is a flowchart illustrating an operation of the vehicle **200** which is a reception side. FIG. **6** is a flowchart illustrating an operation of the vehicle **200** which is the reception side and a diagram continuous from FIG. **5**.

First, an operation of the vehicle **100** will be described in detail with reference to the flowchart of FIG. **4**. In step S**301**, the vehicle **100** starts a process. In step S**302**, the front camera **107** and the rear camera **108** image the front and rear of the vehicle **100**, respectively, and transmit videos to the image processing unit **109**.

In step S**303**, the image processing unit **109** performs image processing such as de-Bayer processing, distortion correction, gamma curve processing, or color space conversion on the received videos of the front camera **107** and the rear camera **108**. In step S**303**, the image processing unit **109** outputs the generated videos to the system control unit **106** and the foreign object recognition unit **110**.

In step S**304**, the foreign object recognition unit **110** performs a recognition process on the received videos, searches for a foreign object on the road, and determines whether there is a foreign object. When the foreign object recognition unit **110** determines that there is the foreign object, the process of step S**305** is performed. When the foreign object recognition unit **110** determines that there is no foreign object, the process returns to step S**302** and continues.

In step S**305**, the system control unit **106** selects a video in which the foreign object is imaged among the videos received from the image processing unit **109** and temporarily stores the video in the data storage unit **104** when the foreign object recognition unit **110** notifies the system control unit **106** that there is the foreign object.

In step S**306**, the system control unit **106** acquires a present time from the present time acquisition unit **101**, acquires a present location from the present location acquisition unit **102**, and acquires an imaging direction from the imaging direction acquisition unit **103**.

In step S**307**, the system control unit **106** adds various types of information in association to the video in which the foreign object on the road is imaged and which is stored in the data storage unit **104**. Examples of the various types of information added in association to the video in which the foreign object is imaged include information in which the present time acquired in step S**306** is set as the imaging time.

The present time acquisition unit **101** is an example of an imaging time acquisition unit. Examples of the various types of information added in association to the video in which the foreign object is imaged include information in which the present location acquired in step S**306** is an imaging location.

The present location acquisition unit **102** is an example of an imaging location acquisition unit. Examples of the various types of information added in association to the video in which the foreign object is imaged include information regarding the imaging direction acquired in step S**306**.

Thereafter, in step S**307**, the system control unit **106** uses the communication unit **105** to transmit, to the outside, a data set of the video of the foreign object on the road, the imaging time, the imaging location, and the imaging direction which are associated.

Here, the outside is a vehicle in the periphery of the present location. The vehicle **100** notifies the vehicle in the periphery of information regarding the foreign object on the road by transmitting the information using a broadcast scheme through inter-vehicle communication.

In step S**308**, the system control unit **106** determines whether a sequence end is notified of by a control signal from the outside (not illustrated). When the system control unit **106** determines that the sequence end is notified of, the process of step S**309** is performed.

When the system control unit **106** determines that the sequence end is not notified of, the process returns to step S**302** and continues. In step S**309**, the system control unit **106** ends the process.

The flowchart illustrated in FIG. **4** has been described above. Here, it is considered that the operation of the flowchart of FIG. **4** is applied to the vehicle **100** in FIG. **3**A. When the vehicle **100** reaches the point K, the vehicle **100** traveling in the lane X from the S direction to the N direction can image the A surface of the foreign object **220** with the front camera **107** and detect the foreign object on the road. FIG. **7**A illustrates a video captured by the front camera **107** when the vehicle **100** reaches the point K.

In the video captured with the front camera **107**, as illustrated in FIG. **7**A, the foreign object is on the right lane of the road and the A surface of the foreign object is imaged. In FIG. **7**A, the left lane is the lane X and the right lane is the lane Y.

In FIG. **7**A, to easily understand that the A surface of the foreign object is imaged, "A" is written on the foreign object. In the video of FIG. **7**A, it is assumed that TK is an imaging time, PK is an imaging location, and DK is an imaging direction. The vehicle **100** transmits a data set of the video of FIG. **7**A, the imaging time TK, the imaging location PK, and the imaging direction DK in association with each other.

When the vehicle **100** reaches the point L, the vehicle **100** can image the B surface of the foreign object **220** with the rear camera **108** and detect the foreign object on the road. FIG. **7**B illustrates a video captured with the rear camera **108** when the vehicle **100** reaches the point K.

In the video captured with the rear camera **108**, as illustrated in FIG. **7**B, a foreign object is on the left lane of the road and the B surface of the foreign object is imaged. In FIG. **7**B, the left lane is the lane X and the right lane is the lane Y.

In FIG. **7**B, to easily understand that the B surface of the foreign object is imaged, "B" is written on the foreign object. In the video of FIG. **7**B, it is assumed that TL is an imaging time, PL is an imaging location, and DL is an imaging direction. The vehicle **100** transmits a data set of the video of FIG. **7**B, the imaging time TL, the imaging location PL, and the imaging direction DL in association with each other.

The imaging direction DK is the N direction and the imaging direction DI is the S direction. The imaging directions DK and DL are directions different from each other by 180°. It can be estimated that the same foreign object **220** is imaged in the video of FIG. **7**A and the video of FIG. **7**B from the traveling direction calculated from a temporal change of the present location of the vehicle **100** and the imaging direction of each off of the front camera **107** and the rear camera **108**.

Therefore, the imaging locations PK and PL are positions of the points K and L. In the present embodiment, the imaging locations PK and PL may be treated as the same location.

Next, an operation of the vehicle **200** will be described in detail with reference to the flowcharts of FIGS. **5** and **6**. In step S**401**, the vehicle **200** starts a process. In step S**402**, the system control unit **206** acquires a present location from the present location acquisition unit **202**.

In step S**403**, the system control unit **206** transmits the acquired present location to the route prediction unit **211**. In step S**403**, the route prediction unit **211** generates a route along which the vehicle **200** is predicted to travel from a history of the received present location, map information stored in the data storage unit **204**, destination information set in a navigation system (not illustrated), and the like.

In step S**403**, the route prediction unit **211** transmits the generated predicted route to the system control unit **206**.

In step S**404**, the system control unit **206** determines whether there is received information in the communication unit **205**. When the system control unit **206** determines that there is no received information, the process of step S**402** is performed. When the system control unit **206** determines that there is the received information, the process of step S**405** is performed.

In step S**405**, the system control unit **206** receives a data set of the video of the foreign object on the road captured by another vehicle, the imaging time, the imaging location, and the imaging direction using the communication unit **205**. In step S**406**, the system control unit **206** selects a video in which there is the imaging location on the predicted route of the vehicle **200** from the received information and stores the data set of the video, the imaging time, the imaging location, and the imaging direction in association in the data storage unit **204**.

In step S**407**, the system control unit **206** determines whether a video of an imaging location closest to the present location is updated from information regarding the present location acquired in step S**402** and the imaging time and the imaging location of the data set stored in step S**406**.

When the system control unit **206** determines that the video is not updated, the process of step S**409** is performed. When the system control unit **206** determines that the video is updated, the process of step S**408** is performed.

In step S**408**, the system control unit **206** acquires the imaging location in the data set associated with the video determined to be updated in step S**407** and estimates a traveling schedule direction at the acquired imaging location from the predicted route generated in step S**403**.

This process is an example of the traveling schedule direction acquisition unit. In step S**408**, the system control unit **206** acquires the imaging direction of the data set associated with the video determined to be updated in step S**407** and compares the acquired imaging direction with the traveling schedule direction estimated in step S**408**.

When the system control unit **206** determines that the imaging direction matches the traveling schedule direction, the process of step S**410** is performed. When the system control unit **206** determines that the imaging direction does not match the traveling schedule direction, the process of step S**409** is performed.

In step S**409**, the system control unit **206** checks the imaging time of the video displayed in the image display unit **212**, compares the imaging time with the present time acquired from the present time acquisition unit **201**, and determines whether the imaging time of the presently displayed video is earlier than a given time.

When the system control unit **206** determines that the imaging time of the presently displayed video is not earlier than the given time, the process of step S**402** is performed. When the system control unit **206** determines that the imaging time of the presently displayed video is earlier than the given time, the process of step S**410** is performed.

In step S**410**, the system control unit **206** selects a video of a latest imaging time at which the imaging location is closest to the present location from the data set stored in step S**406** and displays the video in the image display unit **212**.

In step S**411**, the system control unit **206** determines whether a sequence end is notified of by a control signal from the outside (not illustrated). When the system control unit **206** determines that the sequence end is notified of, the process proceeds to step S**411** and ends. When the system control unit **206** determines that the sequence end is not notified of, the process of step S**402** is performed.

The operation of the vehicle **200** has been described above with reference to the flowcharts of FIGS. **5** and **6**. In step S**408**, whether the imaging direction matches the traveling schedule direction is determined as a condition. However, when the imaging direction and the traveling schedule direction are equal to or less than a given angle (for example, 30°), it may be determined that the imaging direction matches the traveling schedule direction. The image display unit **212** may reverse and display the video when the imaging direction does not match the traveling schedule direction.

Here, it is considered that the operation of the flowcharts described in FIGS. **5** and **6** is applied to the vehicle **200** that has the positional relation illustrated in FIG. **3**B. The vehicle **200** is traveling from the N direction to the S direction and is distant from the foreign object **220**.

As described above, in FIG. **3**A, the vehicle **100** transmits a data set of the video of FIG. **7**A, the imaging time TK, the imaging location PK, and the imaging direction DK at the point K when the vehicle **100** is traveling near the foreign object **220**.

The vehicle **100** transmits the data set of the video of FIG. **7**B, the imaging time TL, the imaging location PL, and the imaging direction DL at the point L. Here, a case in which a video of a foreign object that is located at a position more away than a distance between the vehicle **200** and the foreign object **220** and is located on a road earlier than a given time (for example, 30 seconds) is displayed in the image display unit **212** of the vehicle **200** will be considered.

After the vehicle **100** reaches the point K, the vehicle **100** transmits a data set of the video of FIG. **7**A, the imaging time TK, the imaging location PK, and the imaging direction DK and the vehicle **200** receive the data set. The vehicle **200** receives the data set and updates the video of the closest imaging location. Since the presently displayed video is a video earlier than the given time despite mismatch between the imaging direction and the traveling direction of the own vehicle, the display on the image display unit **212** is updated to the video of FIG. **7**A.

Subsequently, after the vehicle **100** reaches the point L, the vehicle **100** transmits a data set of the video of FIG. **7**B, the imaging time TL, the imaging location PL, and the imaging direction DL, and the vehicle **200** receives the data set. The vehicle **200** receives the data set and updates the video of the closest imaging location because of the match between the imaging direction and the traveling direction of the own vehicle, the display on the image display unit **212** is updated to the video of FIG. **7**B.

The vehicle **200** is a vehicle that travels in the vehicle Y from the N direction to the S direction. Therefore, an ideal video of the foreign object **220** on the road for a driver of the vehicle **200** is a video obtained by imaging the B surface of the foreign object **220** by the vehicle traveling in the lane Y from the N direction to the S direction, as illustrated in FIG.

8. However, when there is no preceding vehicle of the vehicle **200**, the video of FIG. **8** cannot be captured.

When the video of FIG. **7**B is compared with the video of FIG. **8**, the foreign object **220** is located on the left of the screen and the B surface is imaged in both videos. Therefore, FIG. **7**B is a video sufficiently useful for a driver. When the present embodiment is not applied, only the video of FIG. **7**A is provided from the vehicle **100** which is an oncoming vehicle of the vehicle **200**.

The video of FIG. **7**A has right and left reversed disposition of the foreign object **220** and an opposite imaging surface, compared with the video of FIG. **8**. Therefore, it is difficult for the driver to correctly recognize the foreign object **220**.

When there is no preceding vehicle and an oncoming vehicle has only a front camera, a video displayed in the image display unit **212** is not updated.

When the display on the image display unit **212** remains to be the same video for a given time or more, it may be considered that there is a change in the position or state of the foreign object on the road. Therefore, even when an image with a different direction of travel and the imaging direction of the vehicle **200** is provided, the video display of the image display unit **212** may be updated.

When FIG. **7**B is compared with FIG. **8**, it can be understood that a vanishing point of the road is right and left reversed because of a different traveling lane. To solve this problem, for example, image processing may be performed to obtain an image of FIG. **9** by setting the vanishing point of the video of FIG. **7**B near a center.

In the image of FIG. **9**, a process of shifting a horizontal line to the left so that the center line of the road extends in the vertical direction, and thus a movement amount to the left is set to increase when the vehicle moves from the bottom to the top. The image processing enables a driver to less feel a sense of discomfort caused due to a video captured from a different lane. However, in the image processing, as illustrated in FIG. **9**, distortion occurs in the video of the foreign object.

In the present embodiment, the vehicle **100** transmits information in conformity with a broadcast scheme, but is considered that broadcasting is stopped at a time point at which a present location is away from an imaging location by a given distance or more.

Here, a period in which broadcasting continues may be changed depending on a traveling road or a situation. In the case of an imaging location such as a highway where a speed limit is high, the period in which broadcasting continues may be shortened. In the case of a residential street or the like where a speed limit is low, the period may be lengthened.

When it is determined that a large foreign object on a road interferes with traffic, the period in which broadcasting continues may be lengthened. When the foreign object is small, the period may be shortened. When a foreign object is on an oncoming lane, broadcasting may continue to inform many oncoming vehicles of presence of the foreign object for a period longer than in a case where a foreign object is on a traveling lane of the own vehicle.

In the present embodiment, it is assumed that vehicles directly communicate with each other through inter-vehicle communication. The present invention can also be applied even when vehicles communicate with each other via a server.

As described above, according to the present embodiment, when a foreign object on a road is displayed, a video in which an imaging surface of the foreign object is the same in a direction in which right and left dispositions of the foreign object are the same can also be displayed in a video from an oncoming vehicle.

While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation to encompass all such modifications and equivalent structures and functions.

In addition, as a part or the whole of the control according to the embodiments, a computer program realizing the function of the embodiments described above may be supplied to the communication system or the like through a network or various storage media. Then, a computer (or a CPU, an MPU, or the like) of the communication system or the like may be configured to read and execute the program. In such a case, the program and the storage medium storing the program configure the present invention.

The present invention includes inventions implemented using, for example, at least one processor or circuit configured to have functions of the embodiments explained above. A plurality of processors may be used to perform a distribution process.

This application claims the benefit of Japanese Patent Application No. 2023-020471, filed on Feb. 14, 2023, which is hereby incorporated by reference herein in its entirety.

What is claimed is:

1. A communication system comprising:

a first vehicle; and

a second vehicle,

wherein the first vehicle includes

an imaging unit configured to capture a video; and

at least one processor or circuit configured to function as:

a foreign object recognition unit configured to recognize a foreign object on a road from the video,

an imaging location acquisition unit configured to acquire an imaging location which is a location where the video is captured,

an imaging direction acquisition unit configured to acquire an imaging direction which is a direction in which the video is captured, and

a transmission unit configured to transmit information,

wherein the second vehicle includes

at least one processor or circuit configured to function as:

a reception unit configured to receive the information,

a traveling schedule direction acquisition unit configured to acquire a traveling schedule direction of the second vehicle, and

a display unit configured to display a video,

wherein the transmission unit adds an imaging location acquired by the imaging location acquisition unit and an imaging direction acquired by the imaging direction acquisition unit to a video in which the foreign object recognition unit recognizes that there is a foreign object on a road and transmits the video,

wherein the reception unit receives the video in which it is recognized that there is the foreign object, the imaging location, and the imaging direction,

wherein the display unit displays a video in which the traveling schedule direction acquired by the traveling schedule direction acquisition unit matches the received imaging direction,

wherein the at least one processor or circuit in the first vehicle is further configured to function as an imaging time acquisition unit configured to acquire an imaging time which is a time at which the video is captured, wherein the transmission unit adds the imaging time acquired by the imaging time acquisition unit to a video in which the foreign object recognition unit recognizes that there is a foreign object on a road and transmits the video,

wherein the at least one processor or circuit in the second vehicle is further configured to function as a present time acquisition unit configured to acquire a present time,

wherein the reception unit receives the video in which it is recognized that there is a foreign object, the imaging location, the imaging direction, and the imaging time, and

wherein the display unit displays the received video when an imaging time of a presently displayed video is earlier than the present time by a given time or more despite a mismatch between the traveling schedule direction acquired by the traveling schedule direction acquisition unit and the received imaging direction.

2. The communication system according to claim **1**, wherein the imaging unit includes a front imaging unit that images a front side and a rear imaging unit that images a rear side.

3. The communication system according to claim **1**,

wherein the at least one processor or circuit in the second vehicle is further configured to function as a present location acquisition unit configured to acquire a present location of the second vehicle,

wherein the display unit displays a video in which the imaging location is closest to the present location acquired by the present location acquisition unit.

4. A communication system comprising:

a first vehicle; and

a second vehicle,

wherein the first vehicle includes

an imaging unit configured to capture a video; and

at least one processor or circuit configured to function as:

a foreign object recognition unit configured to recognize a foreign object on a road from the video,

an imaging location acquisition unit configured to acquire an imaging location which is a location where the video is captured,

an imaging direction acquisition unit configured to acquire an imaging direction which is a direction in which the video is captured, and

a transmission unit configured to transmit information,

wherein the second vehicle includes

at least one processor or circuit configured to function as:

a reception unit configured to receive the information,

a traveling schedule direction acquisition unit configured to acquire a traveling schedule direction of the second vehicle, and

a display unit configured to display a video,

wherein the transmission unit adds an imaging location acquired by the imaging location acquisition unit and an imaging direction acquired by the imaging direction acquisition unit to a video in which the foreign object recognition unit recognizes that there is a foreign object on a road and transmits the video,

wherein the reception unit receives the video in which it is recognized that there is the foreign object, the imaging location, and the imaging direction,

wherein the display unit displays a video in which the traveling schedule direction acquired by the traveling schedule direction acquisition unit matches the received imaging direction, and

wherein the display unit reverses and displays the video in a case of a mismatch between the traveling schedule direction acquired by the traveling schedule direction acquisition unit and the received imaging direction.

5. The communication system according to claim **4**, wherein the imaging unit includes a front imaging unit that images a front side and a rear imaging unit that images a rear side.

6. The communication system according to claim **4**,

wherein the at least one processor or circuit in the second vehicle is further configured to function as a present location acquisition unit configured to acquire a present location of the second vehicle,

wherein the display unit displays a video in which the imaging location is closest to the present location acquired by the present location acquisition unit.

7. A method of controlling a communication system including first and second vehicles,

wherein the first vehicle captures a video, recognizes a foreign object on a road from the video, acquires an imaging location which is a location where the video is captured, acquires an imaging direction which is a direction in which the video is captured, and transmits information,

wherein the second vehicle receives the information, acquires a traveling schedule direction of the second vehicle, and displays a video, the method comprising:

adding, in the transmitting, an imaging location acquired in the acquiring of the imaging location and an imaging direction acquired in the acquiring of the imaging direction are to a video in which it is recognized that there is a foreign object on a road in the recognizing of the foreign object;

transmitting the video;

receiving, in the video in which it is recognized that there is a foreign object, the imaging location and the imaging direction;

displaying a video in which the traveling schedule direction acquired in the acquiring of the traveling schedule direction matches the received imaging direction; and

reversing and displaying the video in a case of a mismatch between the acquired traveling schedule direction and the received imaging direction.

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